

FOOTBALL LEAGUE MATCH ANALYSIS USING HADOOP MAPREDUCE AND APACHE PIG

Big Data Lab Report

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Course

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1. Problem Statement

The goal of this project is to analyze football league match data using the Hadoop ecosystem — specifically MapReduce (Java) and Apache Pig — in order to extract meaningful insights across several dimensions of the data, including goals scored, match outcomes, league-level comparison, and season-long team performance. Three MapReduce jobs and five Pig scripts were designed, implemented, and executed against the full dataset on a single-node Hadoop cluster.

2. Dataset Description

The dataset used for this project is English Premier League (EPL) and Spanish La Liga match data for the 2024-25 season, sourced from football-data.co.uk (mirrored via datahub.io) — a well-established, freely available source of structured football statistics. Both leagues share an identical schema, split across two files.

Fields

Field	Description
Date	The date the match was played
HomeTeam / AwayTeam	Names of the home and away teams
FTHG / FTAG	Full-time home and away goals
FTR	Full-time result: H (home win), D (draw), A (away win)
HTHG / HTAG / HTR	Half-time equivalents of the above three fields
Referee	Match referee (blank for all La Liga records in this source — see §6)
HomeShots / AwayShots	Total shots taken by each side
HomeShotsTarget / AwayShotsTarget	Shots on target by each side
HomeFouls / AwayFouls	Fouls committed by each side
HomeCorners / AwayCorners	Corners won by each side
HomeYellow / AwayYellow	Yellow cards shown to each side
HomeRed / AwayRed	Red cards shown to each side

Dataset Statistics

Metric	Value
Source files	epl_2425.csv, laliga_2425.csv (comma-separated, header row present)
Total size on disk	56,594 bytes ($\approx$ 55.3 KB)
Total match records	760 (380 EPL + 380 La Liga)
Distinct leagues	2 (English Premier League, Spanish La Liga)
Storage location	HDFS: /bigdatalab2/input (Java tasks); local disk (Pig tasks, run in -x local mode)

Note: The record count of 760 is confirmed independently across all three MapReduce jobs in this report — every job reports Map input records = 762 (760 match rows plus 2 header rows, one per file), and every job's final reduce-side output accounts for exactly 760 valid matches.

3. Summary of Data Analysis

The following MapReduce design patterns are demonstrated across the three Java jobs:

- Aggregation — Task 1 sums goals scored per team from both home and away appearances, using a Combiner to reduce shuffle traffic
- Composite-Key Aggregation — Task 2 counts win/draw/loss outcomes per team by encoding the outcome directly into the reduce key
- Multi-File, Split-Aware Processing — Task 3 detects which league a record belongs to from its input file name via Hadoop's FileSplit, then aggregates per league

The following three MapReduce analyses were implemented:

- Total Goals Per Team: total goals scored by each team, combining home and away appearances
- Match Outcomes Per Team: win / draw / loss counts per team
- League Comparison: average goals per match, EPL vs. La Liga

The following five Pig analyses were implemented:

- High-Scoring Matches: matches with more than 4 total goals
- Total Goals Per Team: the Pig equivalent of MapReduce Task 1
- Top 10 Scoring Teams: teams ranked by total goals scored
- Matches With Country Lookup: JOIN against a derived Team → Country lookup table
- Team Summary: full win/draw/loss, goals for/against, and goal difference per team

4. MapReduce Analysis

All three jobs are implemented as a Driver (main method), Mapper, and Reducer nested as static classes within a single .java file per task — a common, valid Hadoop pattern. Each reads both epl_2425.csv and laliga_2425.csv from a single HDFS input directory (/bigdatalab2/input).

4.1 Task 1: Total Goals Per Team

Returns the total number of goals scored by every team, combining home and away appearances

This analysis comprises of:

- Driver (main method): sets up the job, wiring the Mapper and Reducer together, and reuses the Reducer as the Combiner
- Mapper Class (GoalsMapper): emits <TeamName, Goals> once for the home team and once for the away team of every match
- Reducer Class (GoalsReducer): sums the values for each team; used as both Combiner and final Reducer, which is safe since summing is associative

Mapper Class

```
public static class GoalsMapper extends Mapper<Object, Text, Text, IntWritable> {

    private final Text teamKey = new Text();
    private final IntWritable goals = new IntWritable();

    @Override
    public void map(Object key, Text value, Context context) throws IOException, InterruptedException {
        String line = value.toString().trim();
        if (line.isEmpty() || line.startsWith("Date,") ) {
            return; // skip blank lines and the header row
        }

        String[] fields = line.split(",", -1);
```

```

        if (fields.length < 5) {
            return; // malformed row, skip
        }

        String homeTeam = fields[1].trim();
        String awayTeam = fields[2].trim();

        try {
            int homeGoals = Integer.parseInt(fields[3].trim());
            int awayGoals = Integer.parseInt(fields[4].trim());

            teamKey.set(homeTeam);
            goals.set(homeGoals);
            context.write(teamKey, goals);

            teamKey.set(awayTeam);
            goals.set(awayGoals);
            context.write(teamKey, goals);
        } catch (NumberFormatException e) {
            // skip rows with missing/invalid goal counts
        }
    }
}

```

Reducer Class (also used as Combiner)

```

public static class GoalsReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    private final IntWritable result = new IntWritable();

    @Override
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {
        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }
        result.set(sum);
        context.write(key, result);
    }
}

```

Driver — main() method

```

public static void main(String[] args) throws Exception {
    if (args.length < 2) {
        System.err.println("Usage: TotalGoalsPerTeam <input path> <output path>");
        System.exit(1);
    }

    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Total Goals Per Team");
    job.setJarByClass(TotalGoalsPerTeam.class);
    job.setMapperClass(GoalsMapper.class);
    job.setCombinerClass(GoalsReducer.class);
    job.setReducerClass(GoalsReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}

```

Execution

Command: `hadoop jar TotalGoalsPerTeam.jar TotalGoalsPerTeam /bigdatalab2/input /bigdatalab2/output/task1`

Execution log (excerpt):

```

INFO input.FileInputFormat: Total input files to process : 2
INFO mapreduce.JobSubmitter: number of splits:2
INFO impl.YarnClientImpl: Submitted application application_1788896595952_0002
INFO mapreduce.Job: Running job: job_1788896595952_0002
INFO mapreduce.Job: map 0% reduce 0%
INFO mapreduce.Job: map 100% reduce 0%
INFO mapreduce.Job: map 100% reduce 100%
INFO mapreduce.Job: Job job_1788896595952_0002 completed successfully

```

Job Counters (selected):

Map input records	762
Map output records	1,520
Combine output records	40
Reduce input groups	40
Reduce output records	40
Bytes Written	499

Result

Team	Total Goals
Alaves	38
Arsenal	69
Aston Villa	58
Ath Bilbao	54
Ath Madrid	68
Barcelona	102
Betis	57
Bournemouth	58
Brentford	66
Brighton	66
Celta	59
Chelsea	64
Crystal Palace	51
Espanol	40
Everton	42
Fulham	54
Getafe	34
Girona	44
Ipswich	36
Las Palmas	40
Leganes	39
Leicester	33
Liverpool	86
Mallorca	35
Man City	72
Man United	44
Newcastle	68
Nott'm Forest	58
Osasuna	48

Team	Total Goals
Real Madrid	78
Sevilla	42
Sociedad	35
Southampton	26
Tottenham	64
Valencia	44
Valladolid	26
Vallecano	41
Villarreal	71
West Ham	46
Wolves	54

```

C:\Users\ABRAR\OneDrive\Desktop\BigDataLab2\java-src>hdfs dfs -cat /bigdatalab2/output/task1/part-r-00000
Alaves 38
Arsenal 69
Aston Villa 58
Ath Bilbao 54
Ath Madrid 68
Barcelona 102
Betis 57
Bournemouth 58
Brentford 66
Brighton 66
Celta 59
Chelsea 64
Crystal Palace 51
Espanol 40
Everton 42
Fulham 54
Getafe 34
Girona 44
Ipswich 36
Las Palmas 40
Leganes 39
Leicester 33
Liverpool 86
Mallorca 35
Man City 72
Man United 44
Newcastle 68
Nott'm Forest 58
Osasuna 48
Real Madrid 78
Sevilla 42
Sociedad 35
Southampton 26
Tottenham 64
Valencia 44
Valladolid 26
Vallecano 41
Villarreal 71
West Ham 46
Wolves 54

```

Figure 4.1 — hdfs dfs -cat output for Total Goals Per Team

Note: All 40 team totals sum from exactly 1,520 map output records (760 matches × 2 team-appearances each), matching Combine output records and Reduce output records exactly. Barcelona led all teams with 102 goals, followed by Liverpool (86) and Real Madrid (78).

4.2 Task 2: Match Outcomes Per Team (Win / Draw / Loss)

Returns, per team, how many matches were won, drawn, and lost, from both home and away appearances

This analysis comprises of:

- Driver (main method): sets up the job configuration; reuses the Reducer class as the Combiner
- Mapper Class (OutcomeMapper): translates each match's FTR field into a WIN/DRAW/LOSS outcome for both teams, and emits a composite "TeamName,OUTCOME" key
- Reducer Class (OutcomeReducer): sums the occurrences of each composite key

Mapper Class

```
public static class OutcomeMapper extends Mapper<Object, Text, Text, IntWritable> {

    private static final IntWritable ONE = new IntWritable(1);
    private final Text outKey = new Text();

    @Override
    public void map(Object key, Text value, Context context) throws IOException, InterruptedException {
        String line = value.toString().trim();
        if (line.isEmpty() || line.startsWith("Date,")) {
            return;
        }

        String[] fields = line.split(",", -1);
        if (fields.length < 6) {
            return;
        }

        String homeTeam = fields[1].trim();
        String awayTeam = fields[2].trim();
        String result = fields[5].trim(); // FTR: H, D, or A

        String homeOutcome;
        String awayOutcome;

        switch (result) {
            case "H":
                homeOutcome = "WIN";
                awayOutcome = "LOSS";
                break;
            case "A":
                homeOutcome = "LOSS";
                awayOutcome = "WIN";
                break;
            case "D":
                homeOutcome = "DRAW";
                awayOutcome = "DRAW";
                break;
            default:
                return; // skip rows with an unexpected/missing result code
        }

        outKey.set(homeTeam + "," + homeOutcome);
        context.write(outKey, ONE);

        outKey.set(awayTeam + "," + awayOutcome);
        context.write(outKey, ONE);
    }
}
```

Reducer Class (also used as Combiner)

```
public static class OutcomeReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    private final IntWritable result = new IntWritable();

    @Override
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {
        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }
        result.set(sum);
        context.write(key, result);
    }
}
```

Driver — main() method

```
public static void main(String[] args) throws Exception {
    if (args.length < 2) {
        System.err.println("Usage: MatchOutcomesPerTeam <input path> <output path>");
        System.exit(1);
    }

    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Match Outcomes Per Team");
    job.setJarByClass(MatchOutcomesPerTeam.class);
    job.setMapperClass(OutcomeMapper.class);
    job.setCombinerClass(OutcomeReducer.class);
    job.setReducerClass(OutcomeReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
```

Execution

Command: `hadoop jar MatchOutcomesPerTeam.jar MatchOutcomesPerTeam /bigdatalab2/input /bigdatalab2/output/task2`

Execution log (excerpt):

```
INFO input.FileInputFormat: Total input files to process : 2
INFO mapreduce.JobSubmitter: number of splits:2
INFO impl.YarnClientImpl: Submitted application application_1789048239899_0001
INFO mapreduce.Job: Running job: job_1789048239899_0001
INFO mapreduce.Job: map 0% reduce 0%
INFO mapreduce.Job: map 100% reduce 0%
INFO mapreduce.Job: map 100% reduce 100%
INFO mapreduce.Job: Job job_1789048239899_0001 completed successfully
```

Job Counters (selected):

Map input records	762
Map output records	1,520
Combine output records	120
Reduce input groups	120
Reduce output records	120
Bytes Written	2,015

Result

Team	Wins	Draws	Losses
Alaves	10	12	16
Arsenal	20	14	4
Aston Villa	19	9	10
Ath Bilbao	19	13	6
Ath Madrid	22	10	6
Barcelona	28	4	6
Betis	16	12	10
Bournemouth	15	11	12
Brentford	16	8	14
Brighton	16	13	9
Celta	16	7	15

Team	Wins	Draws	Losses
Chelsea	20	9	9
Crystal Palace	13	14	11
Espanol	11	9	18
Everton	11	15	12
Fulham	15	9	14
Getafe	11	9	18
Girona	11	8	19
Ipswich	4	10	24
Las Palmas	8	8	22
Leganes	9	13	16
Leicester	6	7	25
Liverpool	25	9	4
Mallorca	13	9	16
Man City	21	8	9
Man United	11	9	18
Newcastle	20	6	12
Nott'm Forest	19	8	11
Osasuna	12	16	10
Real Madrid	26	6	6
Sevilla	10	11	17
Sociedad	13	7	18
Southampton	2	6	30
Tottenham	11	5	22
Valencia	11	13	14
Valladolid	4	4	30
Vallecano	13	13	12
Villarreal	20	10	8
West Ham	11	10	17
Wolves	12	6	20

```

Bytes Written=2015
C:\Users\ABRAR\OneDrive\Desktop\BigDataLab2\java-src>hdfs dfs -cat /bigdatalab2/output/task2/part-r-00000
Alaves,DRAW      12
Alaves,LOSS      16
Alaves,WIN       10
Arsenal,DRAW     14
Arsenal,LOSS      4
Arsenal,WIN      20
Aston Villa,DRAW    9
Aston Villa,LOSS   10
Aston Villa,WIN   19
Ath Bilbao,DRAW   13
Ath Bilbao,LOSS    6
Ath Bilbao,WIN    19
Ath Madrid,DRAW   10
Ath Madrid,LOSS    6
Ath Madrid,WIN    22
Barcelona,DRAW     4
Barcelona,LOSS     6
Barcelona,WIN    28
Betis,DRAW        12
Betis,LOSS        10
Betis,WIN         16
Bournemouth,DRAW  11
Bournemouth,LOSS  12
Bournemouth,WIN  15
Brentford,DRAW     8
Brentford,LOSS    14
Brentford,WIN     16
Brighton,DRAW     13
Brighton,LOSS      9
Brighton,WIN      16
Celta,DRAW         7
Celta,LOSS        15
Celta,WIN         16
Chelsea,DRAW       9
Chelsea,LOSS       9
Chelsea,WIN       20

```

Figure 4.2a — Match outcomes per team (part 1)

```

Crystal Palace,DRAW    14
Crystal Palace,LOSS    11
Crystal Palace,WIN     13
Espanol,DRAW          9
Espanol,LOSS          18
Espanol,WIN           11
Everton,DRAW          15
Everton,LOSS          12
Everton,WIN           11
Fulham,DRAW           9
Fulham,LOSS           14
Fulham,WIN            15
Getafe,DRAW           9
Getafe,LOSS           18
Getafe,WIN            11
Girona,DRAW           8
Girona,LOSS           19
Girona,WIN            11
Ipswich,DRAW          10
Ipswich,LOSS          24
Ipswich,WIN           4
Las Palmas,DRAW        8
Las Palmas,LOSS        22
Las Palmas,WIN         8
Leganes,DRAW          13
Leganes,LOSS          16
Leganes,WIN           9
Leicester,DRAW         7
Leicester,LOSS        25
Leicester,WIN         6
Liverpool,DRAW         9
Liverpool,LOSS         4
Liverpool,WIN         25
Mallorca,DRAW          9
Mallorca,LOSS         16
Mallorca,WIN          13
Man City,DRAW          8
Man City,LOSS          9
Man City,WIN          21
Man United,DRAW        9
Man United,LOSS        18
Man United,WIN        11
Newcastle,DRAW         6
Newcastle,LOSS        12
Newcastle,WIN         20

```

Figure 4.2b — Match outcomes per team (part 2)

Note: 120 output records = 40 teams × 3 outcomes each, exactly matching Reduce output records. Every team's three counts sum to 38 (a full season) — e.g. Arsenal recorded 20 wins, 14 draws, and 4 losses.

4.3 Task 3: League Comparison (Average Goals Per Match)

Returns, per league, the match count, total goals, and average goals per match

This analysis comprises of:

- Driver (main method): configures the job; sets Map output types separately from the final output types, since the Reducer's output type (Text) differs from the Mapper's (IntWritable)
- Mapper Class (LeagueMapper): since no column identifies the league directly, the league is detected from the name of the input file the record was read from, using Hadoop's FileSplit
- Reducer Class (LeagueReducer): sums total goals and counts matches per league, then computes the average

Mapper Class

```
public static class LeagueMapper extends Mapper<Object, Text, Text, IntWritable> {

    private final Text leagueKey = new Text();
    private final IntWritable totalGoals = new IntWritable();

    @Override
    public void map(Object key, Text value, Context context) throws IOException, InterruptedException {
        String line = value.toString().trim();
        if (line.isEmpty() || line.startsWith("Date,") ) {
            return;
        }

        String[] fields = line.split(",", -1);
        if (fields.length < 5) {
            return;
        }

        try {
            int homeGoals = Integer.parseInt(fields[3].trim());
            int awayGoals = Integer.parseInt(fields[4].trim());

            FileSplit fileSplit = (FileSplit) context.getInputSplit();
            String fileName = fileSplit.getPath().getName().toLowerCase();
            String league = fileName.contains("laliga") ? "LaLiga" : "EPL";

            leagueKey.set(league);
            totalGoals.set(homeGoals + awayGoals);
            context.write(leagueKey, totalGoals);
        } catch (NumberFormatException e) {
            // skip rows with missing/invalid goal counts
        }
    }
}
```

Reducer Class

```
public static class LeagueReducer extends Reducer<Text, IntWritable, Text, Text> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {
        int totalGoals = 0;
        int matchCount = 0;

        for (IntWritable val : values) {
            totalGoals += val.get();
            matchCount++;
        }

        double average = matchCount == 0 ? 0.0 : (double) totalGoals / matchCount;
        String summary = String.format("Matches=%d, TotalGoals=%d, AvgGoalsPerMatch=%.2f",
            matchCount, totalGoals, average);

        context.write(key, new Text(summary));
    }
}
```

Driver — main() method

```
public static void main(String[] args) throws Exception {
    if (args.length < 2) {
        System.err.println("Usage: LeagueGoalsComparison <input path> <output path>");
        System.exit(1);
    }

    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "League Goals Comparison");
    job.setJarByClass(LeagueGoalsComparison.class);
    job.setMapperClass(LeagueMapper.class);
    job.setReducerClass(LeagueReducer.class);

    job.setMapOutputKeyClass(Text.class);
    job.setMapOutputValueClass(IntWritable.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(Text.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
```

Execution

Command: `hadoop jar LeagueGoalsComparison.jar LeagueGoalsComparison /bigdatalab2/input /bigdatalab2/output/task3`

Execution log (excerpt):

```
INFO input.FileInputFormat: Total input files to process : 2
INFO mapreduce.JobSubmitter: number of splits:2
INFO impl.YarnClientImpl: Submitted application application_1789048239899_0002
INFO mapreduce.Job: Running job: job_1789048239899_0002
INFO mapreduce.Job: map 0% reduce 0%
INFO mapreduce.Job: map 100% reduce 0%
INFO mapreduce.Job: map 100% reduce 100%
INFO mapreduce.Job: Job job_1789048239899_0002 completed successfully
```

Job Counters (selected):

Map input records	762
Map output records	760
Combiner	not used for this task
Reduce input groups	2
Reduce output records	2
Bytes Written	114

Result

League	Matches	Total Goals	Avg. Goals/Match
EPL	380	1,115	2.93
LaLiga	380	995	2.62

```
C:\Users\ABRAR\OneDrive\Desktop\BigDataLab2\java-src>hdfs dfs -cat /bigdatalab2/output/task3/part-r-00000
EPL    Matches=380, TotalGoals=1115, AvgGoalsPerMatch=2.93
LaLiga Matches=380, TotalGoals=995, AvgGoalsPerMatch=2.62
```

Figure 4.3 — `hdfs dfs -cat` output for League Comparison

Note: Map output records (760) equals exactly the number of valid matches, and the two Reduce input groups correspond exactly to the two leagues. The EPL averaged noticeably more goals per match (2.93) than La Liga (2.62) across an identical number of matches (380 each).

5. Pig Analysis

All five Pig scripts were run in local mode (`pig -x local -f <script>`), reading `epl_2425.csv` and `laliga_2425.csv` directly from the `local dataset\` folder.

5.1 Task 1: High-Scoring Matches

Returns all matches with more than 4 total goals, across both leagues

Script (`task1_filter_high_scoring.pig`)

```
-- Task 1: FILTER
-- Loads both leagues, tags each with its League, combines them, and filters
-- down to high-scoring matches (more than 4 total goals).
```

```

epl_raw = LOAD 'dataset/epl_2425.csv' USING PigStorage(',') AS
(Date:chararray, HomeTeam:chararray, AwayTeam:chararray, FTHG:int, FTAG:int, FTR:chararray,
HTHG:int, HTAG:int, HTR:chararray, Referee:chararray, HomeShots:int, AwayShots:int,
HomeShotsTarget:int, AwayShotsTarget:int, HomeFouls:int, AwayFouls:int, HomeCorners:int,
AwayCorners:int, HomeYellow:int, AwayYellow:int, HomeRed:int, AwayRed:int);

laliga_raw = LOAD 'dataset/laliga_2425.csv' USING PigStorage(',') AS
(Date:chararray, HomeTeam:chararray, AwayTeam:chararray, FTHG:int, FTAG:int, FTR:chararray,
HTHG:int, HTAG:int, HTR:chararray, Referee:chararray, HomeShots:int, AwayShots:int,
HomeShotsTarget:int, AwayShotsTarget:int, HomeFouls:int, AwayFouls:int, HomeCorners:int,
AwayCorners:int, HomeYellow:int, AwayYellow:int, HomeRed:int, AwayRed:int);

-- Drop the header row from each file (its Date field is the literal string 'Date')
epl_clean = FILTER epl_raw BY Date != 'Date';
laliga_clean = FILTER laliga_raw BY Date != 'Date';

-- Tag each match with which league it came from
epl_tagged = FOREACH epl_clean GENERATE Date, HomeTeam, AwayTeam, FTHG, FTAG, 'EPL' AS League;
laliga_tagged = FOREACH laliga_clean GENERATE Date, HomeTeam, AwayTeam, FTHG, FTAG, 'LaLiga' AS League;

all_matches = UNION epl_tagged, laliga_tagged;

-- Compute total goals per match, then filter for high-scoring games
with_total = FOREACH all_matches GENERATE Date, HomeTeam, AwayTeam, League, (FTHG + FTAG) AS TotalGoals;

high_scoring = FILTER with_total BY TotalGoals > 4;

STORE high_scoring INTO 'output/pig_task1_high_scoring' USING PigStorage(',');

```

Execution

Command: pig -x local -f pig-scripts/task1_filter_high_scoring.pig

Script executed in local mode and completed with “Success!”, reading 381 lines from each input file (380 matches + 1 header) and storing 109 high-scoring matches to output\pig_task1_high_scoring.

Results

Preview (first 10 of 109 high-scoring matches; full list in Figure 5.1):

Date	HomeTeam	AwayTeam	League	TotalGoals
2024-08-24	Man City	Ipswich	EPL	5
2024-08-25	Wolves	Chelsea	EPL	8
2024-08-31	Everton	Bournemouth	EPL	5
2024-09-14	Aston Villa	Everton	EPL	5
2024-09-28	Arsenal	Leicester	EPL	6
2024-09-28	Chelsea	Brighton	EPL	6
2024-10-05	Brentford	Wolves	EPL	8
2024-10-05	Man City	Fulham	EPL	5
2024-10-05	West Ham	Ipswich	EPL	5
2024-10-06	Brighton	Tottenham	EPL	5

```

C:\Users\ABRAR\OneDrive\Desktop\BigDataLab2>type output\pig_task1_high_scoring\part-m-00000
2024-08-24,Man City,Ipswich,EPL,5
2024-08-25,Wolves,Chelsea,EPL,8
2024-08-31,Everton,Bournemouth,EPL,5
2024-09-14,Aston Villa,Everton,EPL,5
2024-09-28,Arsenal,Leicester,EPL,6
2024-09-28,Chelsea,Brighton,EPL,6
2024-10-05,Brentford,Wolves,EPL,8
2024-10-05,Man City,Fulham,EPL,5
2024-10-05,West Ham,Ipswich,EPL,5
2024-10-06,Brighton,Tottenham,EPL,5
2024-10-19,Tottenham,West Ham,EPL,5
2024-10-19,Southampton,Leicester,EPL,5
2024-10-26,Brentford,Ipswich,EPL,7
2024-11-03,Tottenham,Aston Villa,EPL,5
2024-11-09,Brentford,Bournemouth,EPL,5
2024-11-23,Fulham,Wolves,EPL,5
2024-11-24,Southampton,Liverpool,EPL,5
2024-11-30,Brentford,Leicester,EPL,5
2024-11-30,Wolves,Bournemouth,EPL,6
2024-11-30,West Ham,Arsenal,EPL,7
2024-12-04,Newcastle,Liverpool,EPL,6
2024-12-04,Southampton,Chelsea,EPL,6
2024-12-07,Brentford,Newcastle,EPL,6
2024-12-07,Man United,Nott'm Forest,EPL,5
2024-12-08,Tottenham,Chelsea,EPL,7
2024-12-15,Southampton,Tottenham,EPL,5
2024-12-21,Crystal Palace,Arsenal,EPL,6
2024-12-22,Tottenham,Liverpool,EPL,9
2024-12-29,West Ham,Liverpool,EPL,5
2025-01-04,Man City,West Ham,EPL,5

```

Figure 5.1a — High-scoring matches (part 1)

```

2024-12-07,Man United,Nott'm Forest,EPL,5
2024-12-08,Tottenham,Chelsea,EPL,7
2024-12-15,Southampton,Tottenham,EPL,5
2024-12-21,Crystal Palace,Arsenal,EPL,6
2024-12-22,Tottenham,Liverpool,EPL,9
2024-12-29,West Ham,Liverpool,EPL,5
2025-01-04,Man City,West Ham,EPL,5
2025-01-04,Southampton,Brentford,EPL,5
2025-01-14,West Ham,Fulham,EPL,5
2025-01-18,Newcastle,Bournemouth,EPL,5
2025-01-19,Everton,Tottenham,EPL,5
2025-01-19,Nott'm Forest,Southampton,EPL,5
2025-01-19,Ipswich,Man City,EPL,6
2025-01-25,Bournemouth,Nott'm Forest,EPL,5
2025-01-25,Liverpool,Ipswich,EPL,5
2025-02-01,Nott'm Forest,Brighton,EPL,7
2025-02-02,Arsenal,Man City,EPL,6
2025-02-22,Ipswich,Tottenham,EPL,5
2025-02-23,Newcastle,Nott'm Forest,EPL,7
2025-02-25,Crystal Palace,Aston Villa,EPL,5
2025-02-26,Man United,Ipswich,EPL,5
2025-03-15,Ipswich,Nott'm Forest,EPL,6
2025-04-06,Fulham,Liverpool,EPL,5
2025-04-12,Man City,Crystal Palace,EPL,7
2025-04-13,Wolves,Tottenham,EPL,6
2025-04-13,Newcastle,Man United,EPL,5
2025-04-16,Newcastle,Crystal Palace,EPL,5
2025-04-19,Brentford,Brighton,EPL,6
2025-04-19,Aston Villa,Newcastle,EPL,5
2025-04-26,Brighton,West Ham,EPL,5
2025-04-27,Liverpool,Tottenham,EPL,6
2025-05-04,Brentford,Man United,EPL,7
2025-05-18,Brentford,Fulham,EPL,5
2025-05-19,Brighton,Liverpool,EPL,5
2025-05-20,Crystal Palace,Wolves,EPL,6
2025-05-25,Tottenham,Brighton,EPL,5

```

Figure 5.1b — High-scoring matches (part 2)

Note: 109 of the 760 combined matches (about 14%) had more than four total goals.

5.2 Task 2: Total Goals Per Team

The Pig equivalent of MapReduce Task 1 — total goals scored by each team

Script (task2_total_goals_per_team.pig)

```
-- Task 2: GROUP + aggregate
-- Computes total goals scored by each team, combining their home and away
-- appearances across both leagues.

epl_raw = LOAD 'dataset/epl_2425.csv' USING PigStorage(',') AS
(Date:chararray, HomeTeam:chararray, AwayTeam:chararray, FTHG:int, FTAG:int, FTR:chararray,
HTHG:int, HTAG:int, HTR:chararray, Referee:chararray, HomeShots:int, AwayShots:int,
HomeShotsTarget:int, AwayShotsTarget:int, HomeFouls:int, AwayFouls:int, HomeCorners:int,
AwayCorners:int, HomeYellow:int, AwayYellow:int, HomeRed:int, AwayRed:int);

laliga_raw = LOAD 'dataset/laliga_2425.csv' USING PigStorage(',') AS
(Date:chararray, HomeTeam:chararray, AwayTeam:chararray, FTHG:int, FTAG:int, FTR:chararray,
HTHG:int, HTAG:int, HTR:chararray, Referee:chararray, HomeShots:int, AwayShots:int,
HomeShotsTarget:int, AwayShotsTarget:int, HomeFouls:int, AwayFouls:int, HomeCorners:int,
AwayCorners:int, HomeYellow:int, AwayYellow:int, HomeRed:int, AwayRed:int);

epl_clean = FILTER epl_raw BY Date != 'Date';
laliga_clean = FILTER laliga_raw BY Date != 'Date';

all_matches = UNION epl_clean, laliga_clean;

-- Split into a home-goals view and an away-goals view, then combine them
home_goals = FOREACH all_matches GENERATE HomeTeam AS Team, FTHG AS Goals;
away_goals = FOREACH all_matches GENERATE AwayTeam AS Team, FTAG AS Goals;
all_goals = UNION home_goals, away_goals;

grouped = GROUP all_goals BY Team;
team_totals = FOREACH grouped GENERATE group AS Team, SUM(all_goals.Goals) AS TotalGoals;

STORE team_totals INTO 'output/pig_task2_total_goals' USING PigStorage(',');
```

Execution

Command: pig -x local -f pig-scripts/task2_total_goals_per_team.pig

Script executed in local mode and completed with “Success!”. Output was written to output/pig_task2_total_goals.

Results

Team	Total Goals
Alaves	38
Arsenal	69
Aston Villa	58
Ath Bilbao	54
Ath Madrid	68
Barcelona	102
Betis	57
Bournemouth	58
Brentford	66
Brighton	66
Celta	59
Chelsea	64
Crystal Palace	51
Espanol	40
Everton	42

Team	Total Goals
Fulham	54
Getafe	34
Girona	44
Ipswich	36
Las Palmas	40
Leganes	39
Leicester	33
Liverpool	86
Mallorca	35
Man City	72
Man United	44
Newcastle	68
Nott'm Forest	58
Osasuna	48
Real Madrid	78
Sevilla	42
Sociedad	35
Southampton	26
Tottenham	64
Valencia	44
Valladolid	26
Vallecano	41
Villarreal	71
West Ham	46
Wolves	54

```
C:\Users\ABRAR\OneDrive\Desktop\BigDataLab2>type output\pig_task2_total_goals\part-r-00000
Beti s,57
Celta,59
Alaves,38
Fulham,54
Getafe,34
Girona,44
Wolves,54
Arsenal,69
Chelsea,64
Espanol,40
Everton,42
Ipswich,36
Leganes,39
Osasuna,48
Sevilla,42
Brighton,66
Mallorca,35
Man City,72
Sociedad,35
Valencia,44
West Ham,46
Barcelona,102
Brentford,66
Leicester,33
Liverpool,86
Newcastle,68
Tottenham,64
Vallecano,41
Ath Bilbao,54
Ath Madrid,68
Las Palmas,40
Man United,44
Valladolid,26
Villarreal,71
Aston Villa,58
Bournemouth,58
Real Madrid,78
Southampton,26
Nott'm Forest,58
Crystal Palace,51
```

Figure 5.2 — Total goals per team (Pig)

Note: Results match MapReduce Task 1 exactly — e.g. Barcelona 102, Arsenal 69, Real Madrid 78 — confirming the two independently written pipelines agree.

5.3 Task 3: Top 10 Scoring Teams

Returns the 10 teams with the highest total goals scored

Script (task3_top_scoring_teams.pig)

```
-- Task 3: ORDER
-- Same per-team goal totals as Task 2, but ranked descending to surface the
-- top 10 highest-scoring teams across both leagues.

epl_raw = LOAD 'dataset/epl_2425.csv' USING PigStorage(',') AS
(Date:chararray, HomeTeam:chararray, AwayTeam:chararray, FTHG:int, FTAG:int, FTR:chararray,
HTHG:int, HTAG:int, HTR:chararray, Referee:chararray, HomeShots:int, AwayShots:int,
HomeShotsTarget:int, AwayShotsTarget:int, HomeFouls:int, AwayFouls:int, HomeCorners:int,
AwayCorners:int, HomeYellow:int, AwayYellow:int, HomeRed:int, AwayRed:int);

laliga_raw = LOAD 'dataset/laliga_2425.csv' USING PigStorage(',') AS
(Date:chararray, HomeTeam:chararray, AwayTeam:chararray, FTHG:int, FTAG:int, FTR:chararray,
HTHG:int, HTAG:int, HTR:chararray, Referee:chararray, HomeShots:int, AwayShots:int,
HomeShotsTarget:int, AwayShotsTarget:int, HomeFouls:int, AwayFouls:int, HomeCorners:int,
AwayCorners:int, HomeYellow:int, AwayYellow:int, HomeRed:int, AwayRed:int);

epl_clean = FILTER epl_raw BY Date != 'Date';
laliga_clean = FILTER laliga_raw BY Date != 'Date';

all_matches = UNION epl_clean, laliga_clean;

home_goals = FOREACH all_matches GENERATE HomeTeam AS Team, FTHG AS Goals;
away_goals = FOREACH all_matches GENERATE AwayTeam AS Team, FTAG AS Goals;
all_goals = UNION home_goals, away_goals;

grouped = GROUP all_goals BY Team;
```

```
team_totals = FOREACH grouped GENERATE group AS Team, SUM(all_goals.Goals) AS TotalGoals;

ranked_teams = ORDER team_totals BY TotalGoals DESC;
top10 = LIMIT ranked_teams 10;

STORE top10 INTO 'output/pig_task3_top_scoring_teams' USING PigStorage(',');
```

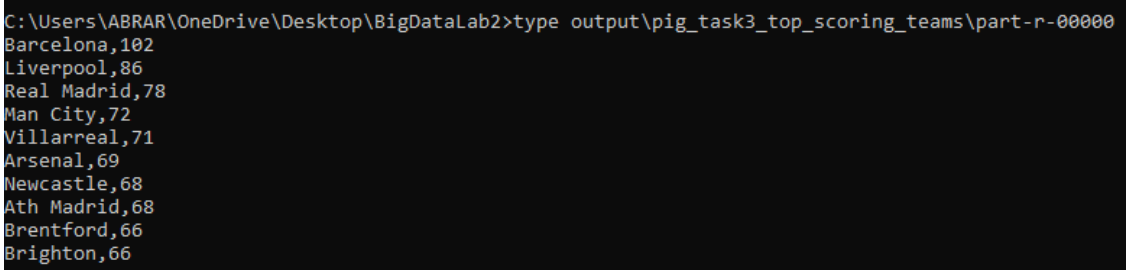
Execution

Command: pig -x local -f pig-scripts\task3_top_scoring_teams.pig

Script executed in local mode and completed with “Success!”. Output was written to output\pig_task3_top_scoring_teams.

Results

Team	Total Goals
Barcelona	102
Liverpool	86
Real Madrid	78
Man City	72
Villarreal	71
Arsenal	69
Newcastle	68
Ath Madrid	68
Brentford	66
Brighton	66



```
C:\Users\ABRAR\OneDrive\Desktop\BigDataLab2>type output\pig_task3_top_scoring_teams\part-r-00000
Barcelona,102
Liverpool,86
Real Madrid,78
Man City,72
Villarreal,71
Arsenal,69
Newcastle,68
Ath Madrid,68
Brentford,66
Brighton,66
```

Figure 5.3 — Top 10 scoring teams

Note: Barcelona tops the list at 102 goals, consistent with both Tasks 1 and 2.

5.4 Task 4: Matches With Country Lookup

Attaches each match's country (England or Spain) via an explicit JOIN

Script (task4_join_team_country.pig)

```
-- Task 4: JOIN
-- Builds a small Team -> League/Country lookup table (derived from the same
-- data, so team-name spelling always matches exactly), then JOINS the match
-- data against it to attach each match's country.

epl_raw = LOAD 'dataset/epl_2425.csv' USING PigStorage(',') AS
(Date:chararray, HomeTeam:chararray, AwayTeam:chararray, FTHG:int, FTAG:int, FTR:chararray,
HTHG:int, HTAG:int, HTR:chararray, Referee:chararray, HomeShots:int, AwayShots:int,
HomeShotsTarget:int, AwayShotsTarget:int, HomeFouls:int, AwayFouls:int, HomeCorners:int,
AwayCorners:int, HomeYellow:int, AwayYellow:int, HomeRed:int, AwayRed:int);

laliga_raw = LOAD 'dataset/laliga_2425.csv' USING PigStorage(',') AS
(Date:chararray, HomeTeam:chararray, AwayTeam:chararray, FTHG:int, FTAG:int, FTR:chararray,
```

```

HTHG:int, HTAG:int, HTR:chararray, Referee:chararray, HomeShots:int, AwayShots:int,
HomeShotsTarget:int, AwayShotsTarget:int, HomeFouls:int, AwayFouls:int, HomeCorners:int,
AwayCorners:int, HomeYellow:int, AwayYellow:int, HomeRed:int, AwayRed:int);

epl_clean = FILTER epl_raw BY Date != 'Date';
laliga_clean = FILTER laliga_raw BY Date != 'Date';

epl_tagged = FOREACH epl_clean GENERATE Date, HomeTeam, AwayTeam, FTHG, FTAG, 'EPL' AS League;
laliga_tagged = FOREACH laliga_clean GENERATE Date, HomeTeam, AwayTeam, FTHG, FTAG, 'LaLiga' AS League;

all_matches = UNION epl_tagged, laliga_tagged;

-- Build the Team -> League/Country lookup table from the same tagged data
epl_teams = FOREACH epl_tagged GENERATE HomeTeam AS TeamName, League;
laliga_teams = FOREACH laliga_tagged GENERATE HomeTeam AS TeamName, League;
all_teams = UNION epl_teams, laliga_teams;
team_league = DISTINCT all_teams;

team_lookup = FOREACH team_league GENERATE TeamName, League,
(League == 'EPL' ? 'England' : 'Spain') AS Country;

-- JOIN each match's home team against the lookup table to attach its country
matches_with_country = JOIN all_matches BY HomeTeam, team_lookup BY TeamName;

result = FOREACH matches_with_country GENERATE
all_matches::Date AS Date,
all_matches::HomeTeam AS HomeTeam,
all_matches::AwayTeam AS AwayTeam,
all_matches::FTHG AS FTHG,
all_matches::FTAG AS FTAG,
team_lookup::Country AS Country;

-- JOIN doesn't guarantee any particular row order, so sort explicitly for a
-- readable, presentable result
sorted_result = ORDER result BY Date;

STORE sorted_result INTO 'output/pig_task4_matches_with_country' USING PigStorage(',');

```

Execution

Command: pig -x local -f pig-scripts/task4_join_team_country.pig

Script executed in local mode and completed with “Success!” (script statistics reported the JOIN feature). Output was written to output/pig_task4_matches_with_country.

Results

Preview (first 10 of 760 matches, chronological order; full list in Figures 5.4a–5.4b):

Date	HomeTeam	AwayTeam	FTHG	FTAG	Country
2024-08-15	Betis	Girona	1	1	Spain
2024-08-15	Ath Bilbao	Getafe	1	1	Spain
2024-08-16	Celta	Alaves	2	1	Spain
2024-08-16	Las Palmas	Sevilla	2	2	Spain
2024-08-16	Man United	Fulham	1	0	England
2024-08-17	Newcastle	Southampton	1	0	England
2024-08-17	Valencia	Barcelona	1	2	Spain
2024-08-17	West Ham	Aston Villa	1	2	England
2024-08-17	Arsenal	Wolves	2	0	England
2024-08-17	Everton	Brighton	0	3	England

```

C:\Users\ABRAR\OneDrive\Desktop\BigDataLab2>type output\pig_task4_matches_with_country\part-r-00000
2024-08-15,Betis,Girona,1,1,Spain
2024-08-15,Ath Bilbao,Getafe,1,1,Spain
2024-08-16,Celta,Alaves,2,1,Spain
2024-08-16,Las Palmas,Sevilla,2,2,Spain
2024-08-16,Man United,Fulham,1,0,England
2024-08-17,Newcastle,Southampton,1,0,England
2024-08-17,Valencia,Barcelona,1,2,Spain
2024-08-17,West Ham,Aston Villa,1,2,England
2024-08-17,Arsenal,Wolves,2,0,England
2024-08-17,Everton,Brighton,0,3,England
2024-08-17,Osasuna,Leganes,1,1,Spain
2024-08-17,Nott'm Forest,Bournemouth,1,1,England
2024-08-17,Ipswich,Liverpool,0,2,England
2024-08-18,Sociedad,Vallecano,1,2,Spain
2024-08-18,Chelsea,Man City,0,2,England
2024-08-18,Mallorca,Real Madrid,1,1,Spain
2024-08-18,Brentford,Crystal Palace,2,1,England
2024-08-19,Valladolid,Espanol,1,0,Spain
2024-08-19,Villarreal,Ath Madrid,2,2,Spain
2024-08-19,Leicester,Tottenham,1,1,England
2024-08-23,Sevilla,Villarreal,1,2,Spain
2024-08-23,Celta,Valencia,3,1,Spain
2024-08-24,Fulham,Leicester,2,1,England
2024-08-24,Southampton,Nott'm Forest,0,1,England
2024-08-24,Brighton,Man United,2,1,England
2024-08-24,Osasuna,Mallorca,1,0,Spain
2024-08-24,Crystal Palace,West Ham,0,2,England
2024-08-24,Aston Villa,Arsenal,0,2,England
2024-08-24,Getafe,Vallecano,0,0,Spain
2024-08-24,Man City,Ipswich,4,1,England
2024-08-24,Tottenham,Everton,4,0,England
2024-08-24,Espanol,Sociedad,0,1,Spain
2024-08-24,Barcelona,Ath Bilbao,2,1,Spain
2024-08-25,Wolves,Chelsea,2,6,England
2024-08-25,Ath Madrid,Girona,3,0,Spain
2024-08-25,Leganes,Las Palmas,2,1,Spain
2024-08-25,Real Madrid,Valladolid,3,0,Spain
2024-08-25,Liverpool,Brentford,2,0,England
2024-08-25,Bournemouth,Newcastle,1,1,England

```

Figure 5.4a — Matches with country attached (part 1)

```

2024-08-25,Liverpool,Brentford,2,0,England
2024-08-25,Bournemouth,Newcastle,1,1,England
2024-08-25,Alaves,Betis,0,0,Spain
2024-08-26,Villarreal,Celta,4,3,Spain
2024-08-27,Mallorca,Sevilla,0,0,Spain
2024-08-27,Vallecano,Barcelona,1,2,Spain
2024-08-28,Valladolid,Leganes,0,0,Spain
2024-08-28,Ath Madrid,Espanol,0,0,Spain
2024-08-28,Sociedad,Alaves,1,2,Spain
2024-08-28,Ath Bilbao,Valencia,1,0,Spain
2024-08-29,Girona,Osasuna,4,0,Spain
2024-08-29,Las Palmas,Real Madrid,1,1,Spain
2024-08-31,Arsenal,Brighton,1,1,England
2024-08-31,Ath Bilbao,Ath Madrid,0,1,Spain
2024-08-31,Valencia,Villarreal,1,1,Spain
2024-08-31,Nott'm Forest,Wolves,1,1,England
2024-08-31,Leganes,Mallorca,0,1,Spain
2024-08-31,Leicester,Aston Villa,1,2,England
2024-08-31,Ipswich,Fulham,1,1,England
2024-08-31,Everton,Bournemouth,2,3,England
2024-08-31,West Ham,Man City,1,3,England
2024-08-31,Espanol,Vallecano,2,1,Spain
2024-08-31,Brentford,Southampton,3,1,England
2024-08-31,Barcelona,Valladolid,7,0,Spain
2024-09-01,Chelsea,Crystal Palace,1,1,England
2024-09-01,Alaves,Las Palmas,2,0,Spain
2024-09-01,Osasuna,Celta,3,2,Spain
2024-09-01,Man United,Liverpool,0,3,England
2024-09-01,Sevilla,Girona,0,2,Spain
2024-09-01,Getafe,Sociedad,0,0,Spain
2024-09-01,Newcastle,Tottenham,2,1,England
2024-09-01,Real Madrid,Betis,2,0,Spain
2024-09-13,Betis,Leganes,2,0,Spain
2024-09-14,Man City,Brentford,2,1,England
2024-09-14,Crystal Palace,Leicester,2,2,England
2024-09-14,Southampton,Man United,0,3,England
2024-09-14,Fulham,West Ham,1,1,England
2024-09-14,Liverpool,Nott'm Forest,0,1,England
2024-09-14,Bournemouth,Chelsea,0,1,England
2024-09-14,Aston Villa,Everton,3,2,England
2024-09-14,Espanol,Alaves,3,2,Spain

```

Figure 5.4b — Matches with country attached (part 2)

Note: A small Team → Country lookup table is built from the same tagged match data (via DISTINCT), so team-name spelling always matches exactly — no external reference file is needed. Because a JOIN's output order is not guaranteed, the result is explicitly sorted by Date.

5.5 Task 5: Team Summary

A full per-team season summary: matches played, wins/draws/losses, goals for/against, and goal difference

Script (task5_team_summary.pig)

```

-- Task 5: GROUP + nested FOREACH
-- Builds a full per-team summary: matches played, wins/draws/losses, goals
-- for/against, and goal difference, then ranks teams by goal difference.

epl_raw = LOAD 'dataset/epl_2425.csv' USING PigStorage(',') AS
(Date:chararray, HomeTeam:chararray, AwayTeam:chararray, FTHG:int, FTAG:int, FTR:chararray,
HTHG:int, HTAG:int, HTR:chararray, Referee:chararray, HomeShots:int, AwayShots:int,
HomeShotsTarget:int, AwayShotsTarget:int, HomeFouls:int, AwayFouls:int, HomeCorners:int,
AwayCorners:int, HomeYellow:int, AwayYellow:int, HomeRed:int, AwayRed:int);

laliga_raw = LOAD 'dataset/laliga_2425.csv' USING PigStorage(',') AS
(Date:chararray, HomeTeam:chararray, AwayTeam:chararray, FTHG:int, FTAG:int, FTR:chararray,
HTHG:int, HTAG:int, HTR:chararray, Referee:chararray, HomeShots:int, AwayShots:int,
HomeShotsTarget:int, AwayShotsTarget:int, HomeFouls:int, AwayFouls:int, HomeCorners:int,
AwayCorners:int, HomeYellow:int, AwayYellow:int, HomeRed:int, AwayRed:int);

epl_clean = FILTER epl_raw BY Date != 'Date';
laliga_clean = FILTER laliga_raw BY Date != 'Date';

all_matches = UNION epl_clean, laliga_clean;

```

```

-- Home-team perspective: goals for/against and the result from that team's view
home_perspective = FOREACH all_matches GENERATE
  HomeTeam AS Team, FTHG AS GoalsFor, FTAG AS GoalsAgainst,
  (FTR == 'H' ? 'WIN' : (FTR == 'D' ? 'DRAW' : 'LOSS')) AS Result;

-- Away-team perspective: same idea, mirrored
away_perspective = FOREACH all_matches GENERATE
  AwayTeam AS Team, FTAG AS GoalsFor, FTHG AS GoalsAgainst,
  (FTR == 'A' ? 'WIN' : (FTR == 'D' ? 'DRAW' : 'LOSS')) AS Result;

team_matches = UNION home_perspective, away_perspective;

grouped_teams = GROUP team_matches BY Team;

-- Nested FOREACH: for each team, filter its own matches into win/draw/loss
-- sub-relations and count/sum across all of them in one pass
team_summary = FOREACH grouped_teams {
  wins = FILTER team_matches BY Result == 'WIN';
  draws = FILTER team_matches BY Result == 'DRAW';
  losses = FILTER team_matches BY Result == 'LOSS';
  GENERATE
    group AS Team,
    COUNT(team_matches) AS MatchesPlayed,
    COUNT(wins) AS Wins,
    COUNT(draws) AS Draws,
    COUNT(losses) AS Losses,
    SUM(team_matches.GoalsFor) AS GoalsFor,
    SUM(team_matches.GoalsAgainst) AS GoalsAgainst,
    (SUM(team_matches.GoalsFor) - SUM(team_matches.GoalsAgainst)) AS GoalDifference;
};

final_summary = ORDER team_summary BY GoalDifference DESC;

STORE final_summary INTO 'output/pig_task5_team_summary' USING PigStorage(',');

```

Execution

Command: pig -x local -f pig-scripts/task5_team_summary.pig

Script executed in local mode and completed with “Success!”. Output was written to output/pig_task5_team_summary.

Results

Team	MP	W	D	L	GF	GA	GD
Barcelona	38	28	4	6	102	39	63
Liverpool	38	25	9	4	86	41	45
Real Madrid	38	26	6	6	78	38	40
Ath Madrid	38	22	10	6	68	30	38
Arsenal	38	20	14	4	69	34	35
Man City	38	21	8	9	72	44	28
Ath Bilbao	38	19	13	6	54	29	25
Chelsea	38	20	9	9	64	43	21
Newcastle	38	20	6	12	68	47	21
Villarreal	38	20	10	8	71	51	20
Bournemouth	38	15	11	12	58	46	12
Nott'm Forest	38	19	8	11	58	46	12
Brentford	38	16	8	14	66	57	9
Brighton	38	16	13	9	66	59	7
Aston Villa	38	19	9	10	58	51	7
Betis	38	16	12	10	57	50	7

Team	MP	W	D	L	GF	GA	GD
Celta	38	16	7	15	59	57	2
Crystal Palace	38	13	14	11	51	51	0
Fulham	38	15	9	14	54	54	0
Tottenham	38	11	5	22	64	65	-1
Everton	38	11	15	12	42	44	-2
Vallecano	38	13	13	12	41	45	-4
Osasuna	38	12	16	10	48	52	-4
Getafe	38	11	9	18	34	39	-5
Mallorca	38	13	9	16	35	44	-9
Alaves	38	10	12	16	38	48	-10
Man United	38	11	9	18	44	54	-10
Valencia	38	11	13	14	44	54	-10
Espanol	38	11	9	18	40	51	-11
Sociedad	38	13	7	18	35	46	-11
Sevilla	38	10	11	17	42	55	-13
Wolves	38	12	6	20	54	69	-15
West Ham	38	11	10	17	46	62	-16
Girona	38	11	8	19	44	60	-16
Leganes	38	9	13	16	39	56	-17
Las Palmas	38	8	8	22	40	61	-21
Ipswich	38	4	10	24	36	82	-46
Leicester	38	6	7	25	33	80	-47
Southampton	38	2	6	30	26	86	-60
Valladolid	38	4	4	30	26	90	-64

```

C:\Users\ABRAR\OneDrive\Desktop\BigDataLab2>type output\pig_task5_team_summary\part-r-00000
Barcelona,38,28,4,6,102,39,63
Liverpool,38,25,9,4,86,41,45
Real Madrid,38,26,6,6,78,38,40
Ath Madrid,38,22,10,6,68,30,38
Arsenal,38,20,14,4,69,34,35
Man City,38,21,8,9,72,44,28
Ath Bilbao,38,19,13,6,54,29,25
Chelsea,38,20,9,9,64,43,21
Newcastle,38,20,6,12,68,47,21
Villarreal,38,20,10,8,71,51,20
Bournemouth,38,15,11,12,58,46,12
Nott'm Forest,38,19,8,11,58,46,12
Brentford,38,16,8,14,66,57,9
Brighton,38,16,13,9,66,59,7
Aston Villa,38,19,9,10,58,51,7
Betis,38,16,12,10,57,50,7
Celta,38,16,7,15,59,57,2
Crystal Palace,38,13,14,11,51,51,0
Fulham,38,15,9,14,54,54,0
Tottenham,38,11,5,22,64,65,-1
Everton,38,11,15,12,42,44,-2
Vallecano,38,13,13,12,41,45,-4
Osasuna,38,12,16,10,48,52,-4
Getafe,38,11,9,18,34,39,-5
Mallorca,38,13,9,16,35,44,-9
Alaves,38,10,12,16,38,48,-10
Man United,38,11,9,18,44,54,-10
Valencia,38,11,13,14,44,54,-10
Espanol,38,11,9,18,40,51,-11
Sociedad,38,13,7,18,35,46,-11
Sevilla,38,10,11,17,42,55,-13
Wolves,38,12,6,20,54,69,-15
West Ham,38,11,10,17,46,62,-16
Girona,38,11,8,19,44,60,-16
Leganes,38,9,13,16,39,56,-17
Las Palmas,38,8,8,22,40,61,-21
Ipswich,38,4,10,24,36,82,-46
Leicester,38,6,7,25,33,80,-47
Southampton,38,2,6,30,26,86,-60
Valladolid,38,4,4,30,26,90,-64

```

Figure 5.5 — Full team summary, sorted by goal difference

Note: Barcelona finished with the best goal difference (+63) from 28 wins, 4 draws, and 6 losses; Valladolid finished with the worst (−64) from just 4 wins across 38 matches.

6. Data Quality Notes

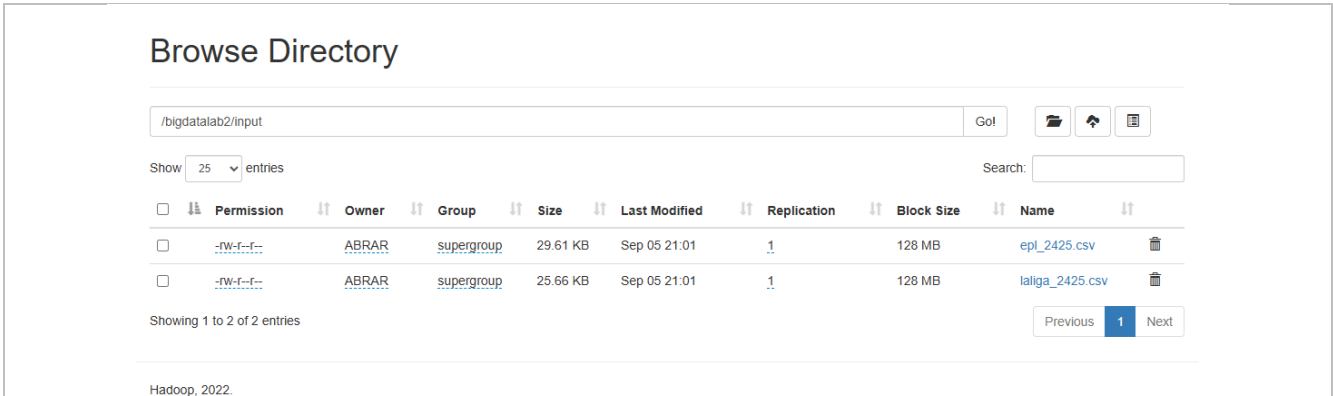
A few characteristics of the raw dataset are worth noting, since they are relevant to interpreting the results above:

- Referee field: the Referee column is blank for every La Liga record in this data source — a known limitation of the underlying provider, not a processing error. None of the eight tasks in this report use the Referee field, so this does not affect any result shown.
- Header rows: each CSV's header row is counted once by Hadoop's FileInputFormat before being explicitly filtered out, both in the Java Mappers (checking for a line starting with "Date,") and in the Pig scripts (FILTER ... BY Date != 'Date'). This is why every Java job's Map input records (762) is exactly two more than the 760 valid match records that reach the output.
- Malformed rows: all three Java Mappers defensively skip rows with fewer fields than expected or a non-numeric goal value, though no such rows were actually encountered in this clean, curated dataset.

7. HDFS Storage Verification

Since the Java tasks were run through Hadoop/YARN, their outputs live natively on HDFS. The Pig tasks, run in local mode, were additionally uploaded to HDFS afterward (`hdfs dfs -put`) purely for verification, so that all eight tasks' results could be confirmed through Hadoop's HDFS web interface (<http://localhost:9870>) in one place.

Input Directory — `/bigdatalab2/input`



Browse Directory

/bigdatalab2/input Go!

Show 25 entries Search:

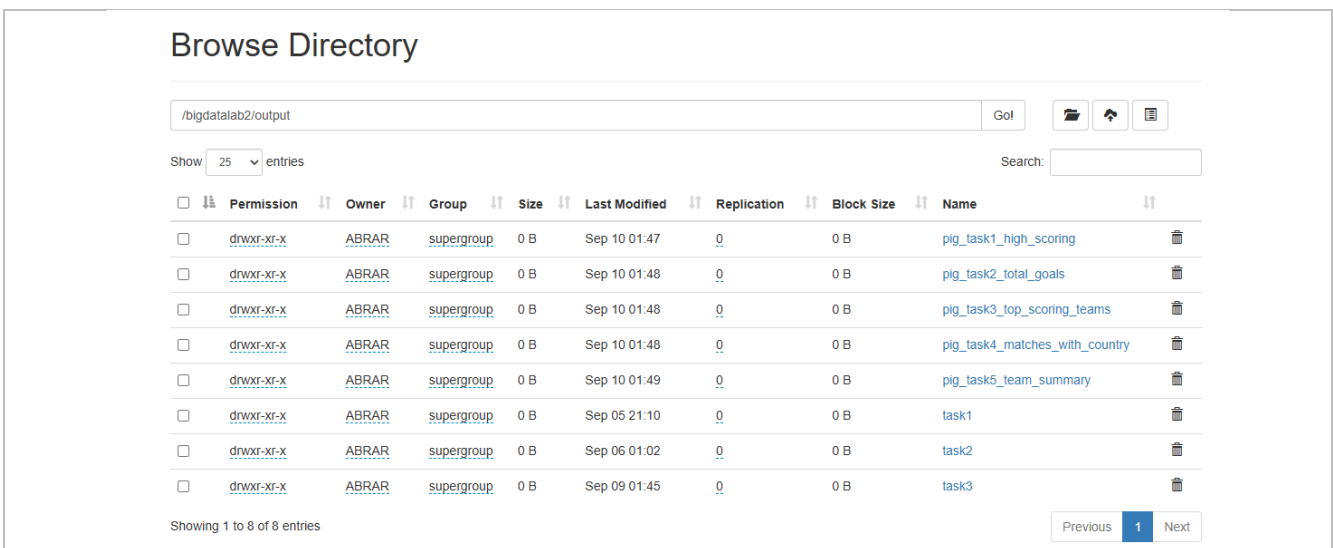
Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
-rw-r--r--	ABRAR	supergroup	29.61 KB	Sep 05 21:01	1	128 MB	epl_2425.csv
-rw-r--r--	ABRAR	supergroup	25.66 KB	Sep 05 21:01	1	128 MB	laliga_2425.csv

Showing 1 to 2 of 2 entries Previous 1 Next

Hadoop, 2022.

Figure 7.1 — Dataset files stored on HDFS

Output Directory — `/bigdatalab2/output`



Browse Directory

/bigdatalab2/output Go!

Show 25 entries Search:

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
drwxr-xr-x	ABRAR	supergroup	0 B	Sep 10 01:47	0	0 B	pig_task1_high_scoring
drwxr-xr-x	ABRAR	supergroup	0 B	Sep 10 01:48	0	0 B	pig_task2_total_goals
drwxr-xr-x	ABRAR	supergroup	0 B	Sep 10 01:48	0	0 B	pig_task3_top_scoring_teams
drwxr-xr-x	ABRAR	supergroup	0 B	Sep 10 01:48	0	0 B	pig_task4_matches_with_country
drwxr-xr-x	ABRAR	supergroup	0 B	Sep 10 01:49	0	0 B	pig_task5_team_summary
drwxr-xr-x	ABRAR	supergroup	0 B	Sep 05 21:10	0	0 B	task1
drwxr-xr-x	ABRAR	supergroup	0 B	Sep 06 01:02	0	0 B	task2
drwxr-xr-x	ABRAR	supergroup	0 B	Sep 09 01:45	0	0 B	task3

Showing 1 to 8 of 8 entries Previous 1 Next

Figure 7.2 — All 8 task outputs verified on HDFS

8. GitHub Repository

The complete project — dataset, Java source and compiled jars, Pig scripts, all eight task outputs, the report and a README — is version-controlled and publicly available at:

github.com/abrarfaisal2286/CSE4346-BigData-Lab2

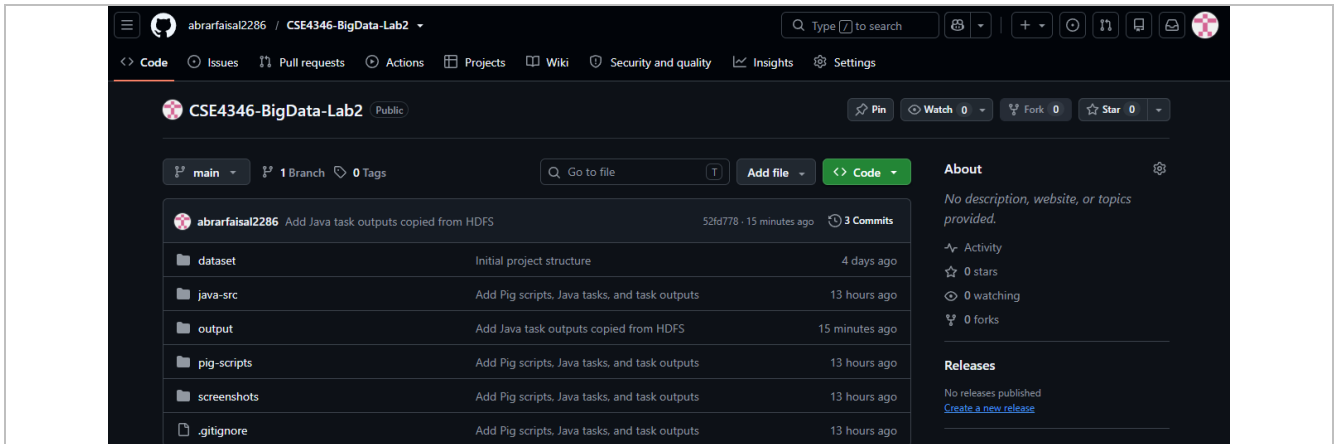


Figure 8.1 — GitHub repository, showing the full commit history and project structure

9. Conclusion

All three MapReduce jobs and all five Pig scripts were implemented, compiled/executed, and verified against the 760-match combined EPL / La Liga 2024-25 dataset on a single-node Hadoop cluster. Cross-checks between independently written jobs — total goals per team matching exactly between the Java and Pig implementations, and Task 3's own printed statistics (EPL: 380 matches, 1,115 goals, 2.93 average; La Liga: 380 matches, 995 goals, 2.62 average) — confirm that both pipelines read and aggregate the dataset consistently. The complete project was verified in two places — the local filesystem and HDFS — and version-controlled on a public GitHub repository.